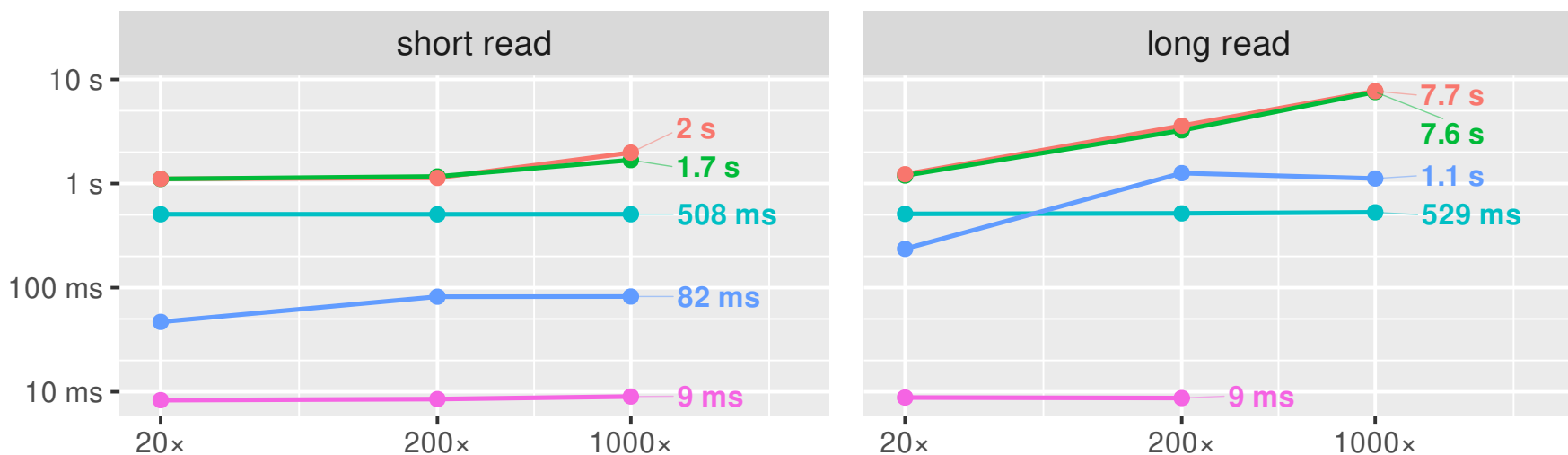
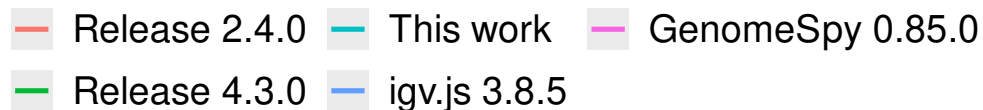
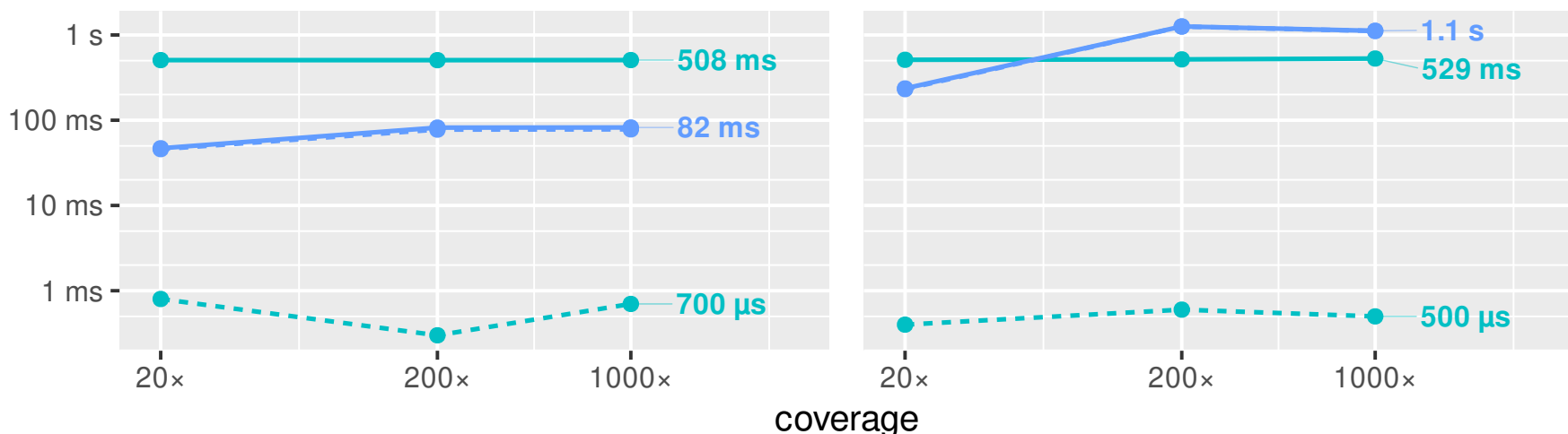


What the user waits for



Inside that wait, for the two arms drawing on the main thread



2× zoom in, median of five steps, timed by one instrument belonging to no tool on the figure. No arm made a network request on any step, so every difference here is drawing and not fetching. Below, igv.js's two lines coincide — its wait is its redraw — while this work spends half a second in a navigation debounce around 0.6 ms of drawing. The two release arms rasterize in a worker, where a page-side clock times the compositing blit and not the rendering, so the lower panel leaves them out. GenomeSpy's long-read line stops at 200×: no 1000× step on that arm returned a usable measurement.